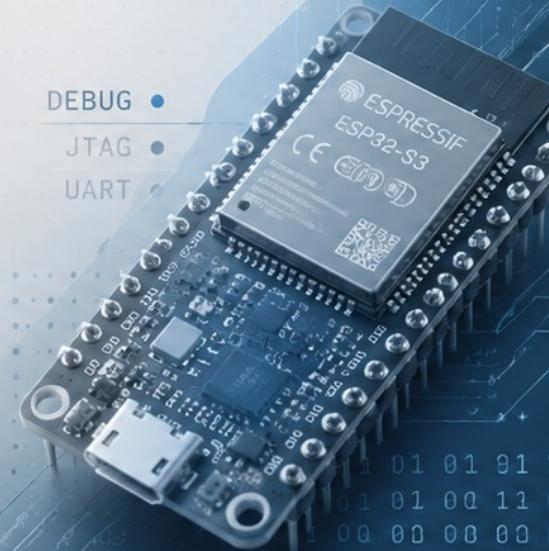


Hardware Debugging met JTAG op de Esp32-S3

DEBUG ●
JTAG ●
UART ●



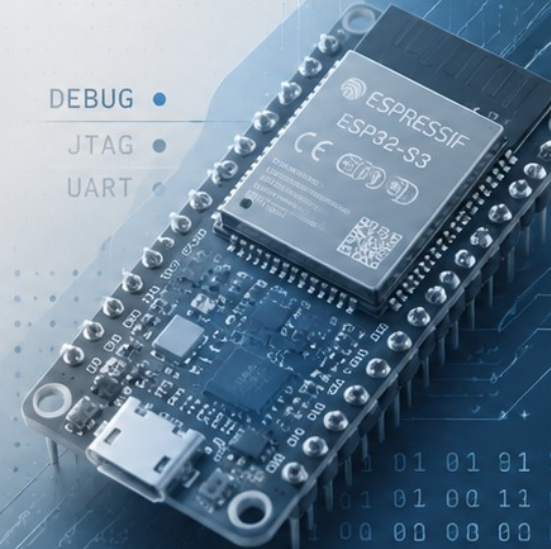
Waarom hardware debugging?

Wie gebruikt hier al een ESP32?

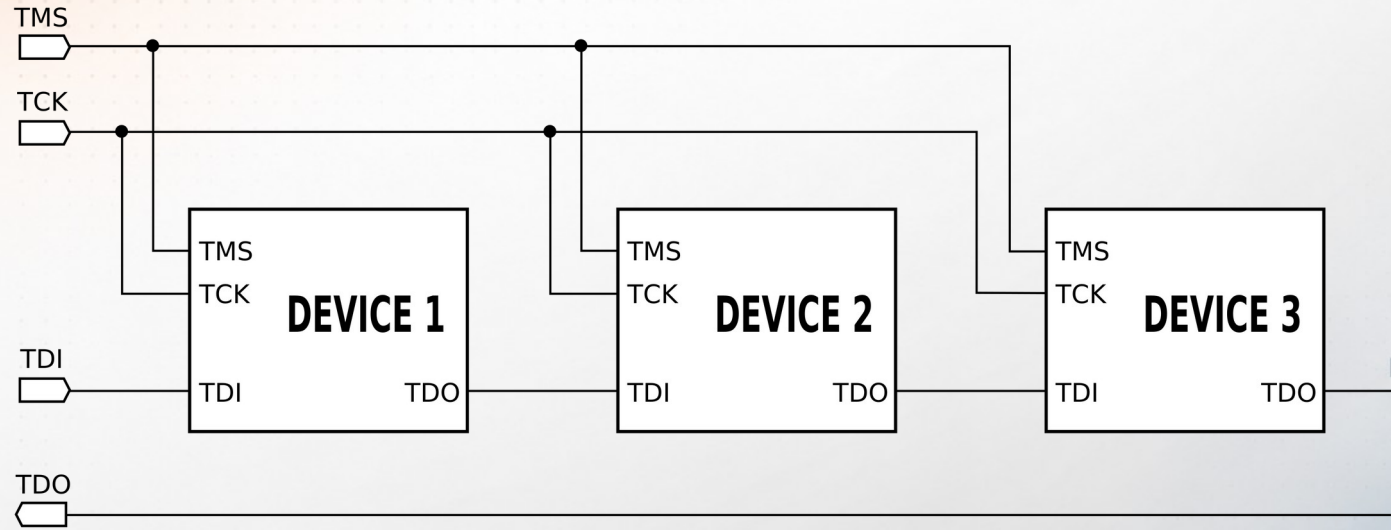
Wie debugt met `Serial.println()`?

Waarom dat soms niet genoeg is

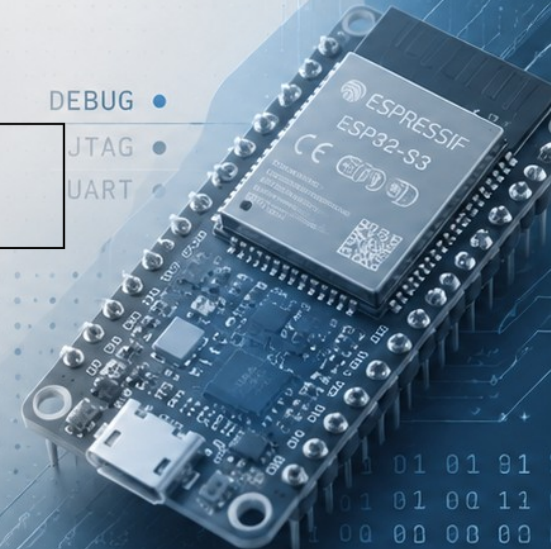
DEBUG ●
JTAG ●
UART ●



Wat is JTAG ?



- DEBUG
- JTAG
- UART

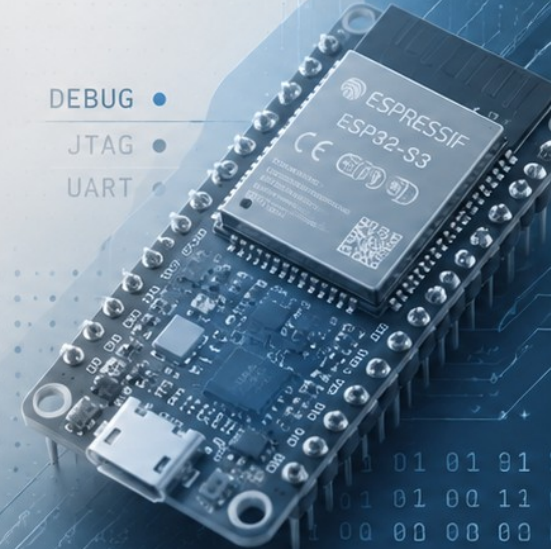


ESP32-S3: wat maakt hem anders?

De ESP32-S3 heeft:

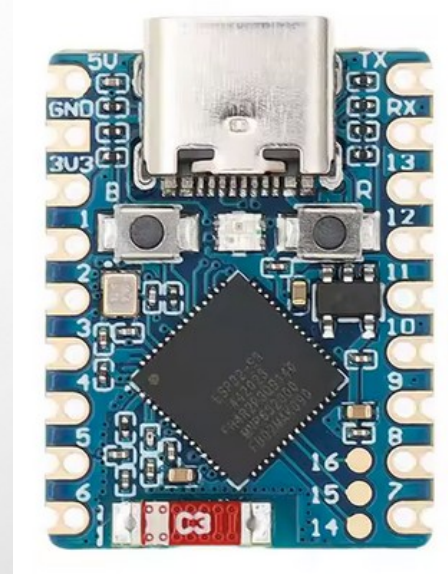
- ✓ **USB-Serial**
- ✓ **USB-JTAG**
- ✓ **Geen extra debugger nodig**
- ✓ **1 kabel = voeding + debug**

DEBUG ●
JTAG ●
UART ●

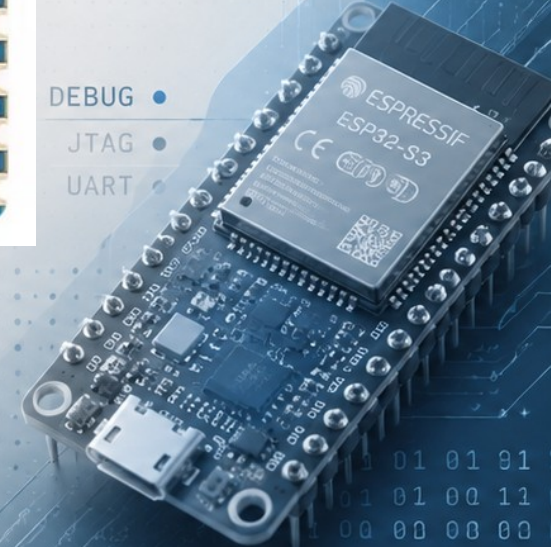


Esp32-S3-Zero

- Chip | ESP32-S3FH4R2
- CPU | Xtensa 32-bit LX7 dual-core
- Clock Speed | Up to 240 MHz
- SRAM | 512 KB
- Flash | 4 MB (8 MB variant available)
- PSRAM | 2 MB (8 MB variant available)
- Wi-Fi | 2.4 GHz 802.11 b/g/n
- Bluetooth | BT 5.0 LE
- USB | Type-C (native, no UART chip)
- GPIO | 24 pins I2C / SPI / UART | 2x / 4x / 3x ADC | 2x 12-bit (up to 20 ch)
- RGB LED | WS2812 on GPIO21 Input Voltage | 3.7V - 6V
- Board Size | 23.5 x 18 mm



DEBUG •
JTAG •
UART •



Hardware debugging mogelijkheden

Breakpoints

Programma pauzeren

CPU staat letterlijk stil

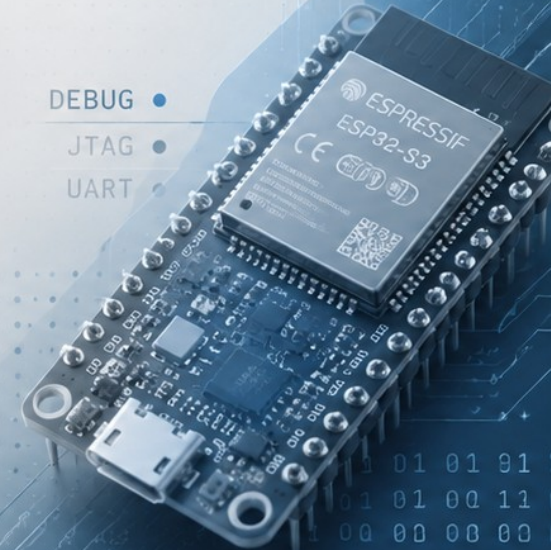
Variabelen bekijken

Sensorwaarden

Structs

Arrays

DEBUG ●
JTAG ●
UART ●



Hardware debugging mogelijkheden

🧠 Call stack

Wie riep wie aan?

Waar zit de crash?

🕒 Step-by-step

Step into

Step over

Step out

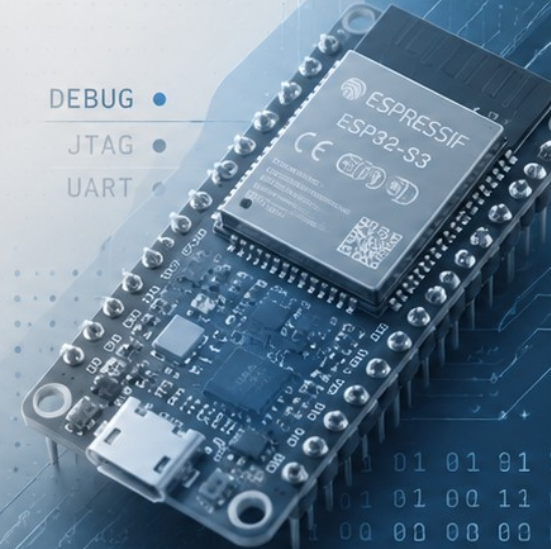
🧵 FreeRTOS awareness

Taken bekijken

Stack gebruik

Task switching

DEBUG ●
JTAG ●
UART ●



De praktijk

Tools:

Arduino IDE <-----

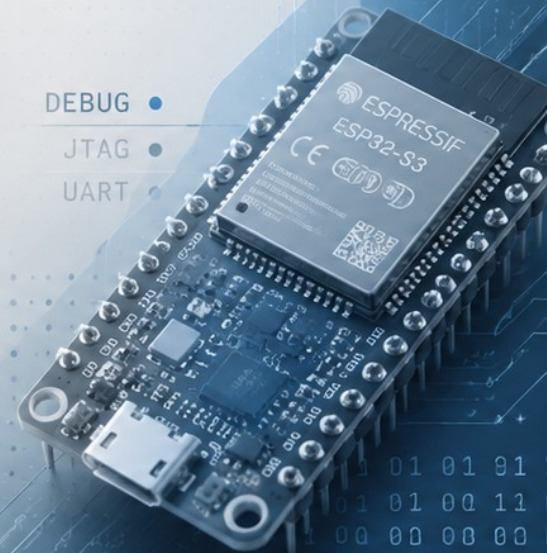
VS Code + PlatformIO plugin

ESP-IDF

DEBUG ●

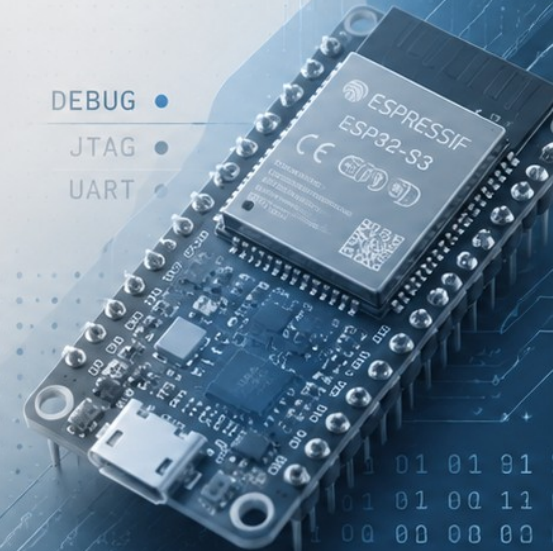
JTAG ●

UART ●



Arduino versie

- Nieuwste versie ≥ 2.3 : <https://www.arduino.cc/en/software/>
- Kies board: ...
- Compileer met debug vlag -g
- Board definitions en selectie

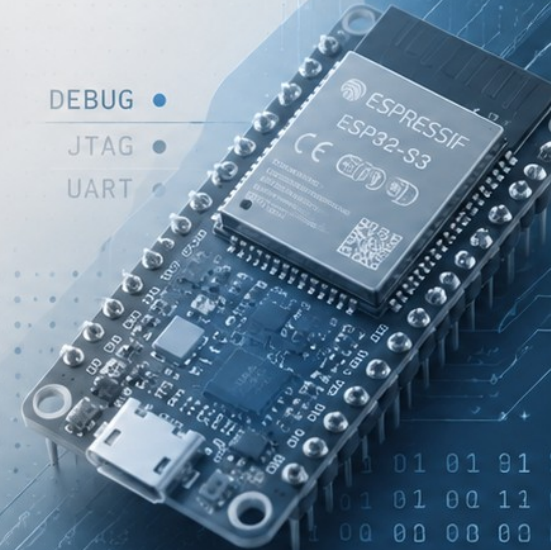


Valkuilen & tips

Tips:

- ✓ Niet tegelijk JTAG debug en serieel op dezelfde kabel, dit werkt niet.
Gebruik aparte serieel kabel op andere pinnen.
- ✓ Gebruik debug builds (compileer met de -g vlag)
- ✓ Zet motor drivers uit tijdens debug
- ✓ Gebruik logging + debugging samen

DEBUG ●
JTAG ●
UART ●



Afsluiting & takeaways

ESP32-S3 maakt professioneel debuggen toegankelijk

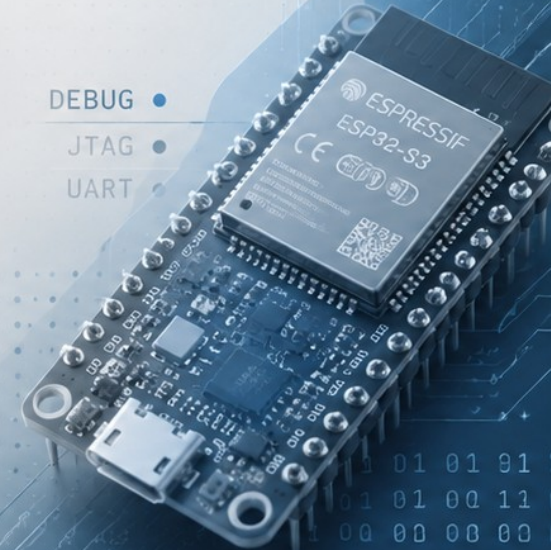
Geen extra hardware nodig

USB ≠ alleen serial

Hardware debuggen bespaart *heel veel tijd*

Minder gokken, meer weten

DEBUG ●
JTAG ●
UART ●



Aanvullende bronnen

- De broncode van de esp32s3-demo :
https://github.com/rouderaa/esp32s3_jtag_demo
- Deze presentatie:
https://github.com/rouderaa/esp32s3_jtag
- Handleiding:
https://github.com/rouderaa/esp32s3_jtag
- Rob vd Ouderaa
rob@bitbouwer.nl

